Joshua Wrigley

Curriculum Vitae

josh.l.wrigley@gmail.com • jlwrigley.github.io
Building Sophie Germain, 8 place Aurélie Nemours, 75013 Paris

Appointments

Jan.–Dec. 2025 **Université Paris Cité** – Paris, France
Postdoctoral Researcher,
Employed on the project: *Topology for Types and Terms*.
Principal investigator: Prof. Sam van Gool.

Apr.–Dec. 2024 **Queen Mary University of London** – London, UK
Postdoctoral Researcher,
Principal investigator: Ivan Tomašić.

Grants

2024 – 2025 **Research Co-Lead:** *Toward a Cohomology of Theories*,
EPSRC Mathematical sciences small grant (£80,860),
Principal investigator: Ivan Tomašić.

Authored the vision and approach sections of the successful submission.

Education

2020 – 2024 **Università degli Studi dell'Insubria** – Como, Italy
PhD in Mathematics and Computer Science,
Thesis: *Doctrinal and Groupoidal Representations of Classifying Topoi*,
Awarded *cum laude*,
Supervisor: Prof. Olivia Caramello.

2019 – 2020 **Lincoln College, University of Oxford** – Oxford, UK
Masters in Mathematics (MMath), Classification: First class,
Dissertation: *Aspects of Stone Duality for Boolean Algebras*,
Supervisor: Prof. Hilary Priestley.

2016 – 2019 **Lincoln College, University of Oxford** – Oxford, UK
Bachelor of Mathematics (BA), Classification: First class.

Publications

2026 **On Topological Groupoids that Represent Theories,**
Zeitschrift für Mathematische Logik und Grundlagen der Mathematik, to appear (2026) (link).

2025 **Day algebras**, with Edmund Robinson,
Accepted in *Mathematical Structures in Computer Science* (preprint: arXiv:2504.06200)

2025 **Topoi with enough points and topological groupoids,**
Journal of Pure and Applied Algebra, Vol. 229(10), 2025, 108073 (link).

2024 **Some Properties of Internal Locale Morphisms Externalised,**
Theory and Applications of Categories, Vol. 41, 2024, No. 35, pp 1160-1202 (link).

2024 **The Representing Localic Groupoid for a Geometric Theory**, with Graham Manuell,
Expositions in Theory and Applications of Categories, 2024, No. 2, pp 1-41 (link).

Preprints

2025 **Existential completions and Herbrand's theorem**, arXiv:2508.15518 (27 pages).

2025 **On the theories classified by an étendue**, arXiv:2507.04526 (17 pages).

2025 **Ultracategories via Kan extensions of relative monads**, with Umberto Tarantino, arXiv:2506.09788 (41 pages).

2024 **Existentially closed models and locally zero-dimensional toposes,** with Mark Kamsma, arXiv:2406.02788 (30 pages).

2023 **The Geometric Completion of a Doctrine,** arXiv:2304.07539 (63 pages, under second review in *Annals of Pure and Applied Logic*).

Teaching experience

2025 **Graduate teaching** (Universitè Paris Citè)
Duality and categorical logic – 12hrs (course lecturer: Sam van Gool).
Lectured exercise classes, including assisting with course design and preparing problem sheets.

2022 – 2024 **Graduate teaching** (Università degli Studi dell'Insubria)
• *Topics in category theory* – 36 hrs (course lecturer: Prof. Olivia Caramello),
• *Topos theory* – 36 hrs (course lecturer: Prof. Olivia Caramello).
Lectured exercise classes and prepared corresponding lecture notes and problem sheets.

2021 – 2022 **Undergraduate teaching** (Università degli Studi dell'Insubria)
• *Algebra 1* – 12 hrs (course lecturer: Prof. Valerio Monti),
• *Algebra 2* – 12 hrs (course lecturer: Prof. Valerio Monti),
• *Geometria 1* – 12 hrs (course lecturer: Prof. Olivia Caramello).
Prepared and taught exam revision lessons, including preparation of problem sheets. Taught in Italian.

Invited talks

April 2026 **TBA.** 7th Workshop on Formal Topology, Venice Italy.

Dec. 2024 **Theories, topological groupoids and topoi.**
Categorical Logic and Higher Categories, University of Manchester, UK.

Professional activities

Dec. 2025 **Scientific committee**
6th ItaCa Workshop, Milan, Italy.

June 2025 **PhD examiner**
Kristóf Kanalas (Masaryk University, Czechia).

Journal referee
Theory and Applications of Categories; *Applied Categorical Structures*.

2024 **Undergraduate admissions interviewer**
Natural sciences, St Edmund's college, Cambridge.

Sept. 2022 **Local Organiser**
9th Symposium on Compositional Structures, Como, Italy.